

Design&Simulation of a CMOS Common-Source Amp

OBJECTIVE:-

The objective of this project was to design and simulate a CMOS common-source amplifier using LTspice, bias it in saturation region, analyze DC operating point, small-signal parameters, frequency response, and noise performance.

TOOLS USED :-

- LTspice XVII
- MOSFET level-1/BSIM model
- AC, DC and Noise analysis
- Parametric sweeps

CIRCUIT DESCRIPTION :-

The circuit consists of an NMOS transistor configured in a common-source topology with a resistive drain load.

A gate biasing network is used to set the operating point of the transistor in the saturation region.

Coupling capacitors are placed at the input and output to block DC components while allowing AC signals to pass.

The design ensures stable amplification with minimal distortion.

The amplifier was designed to operate at a drain current of approximately 70 μA .

The gate-to-source voltage was set near 0.72 V to maintain saturation operation.

The drain-to-source voltage was maintained around 1.07 V.

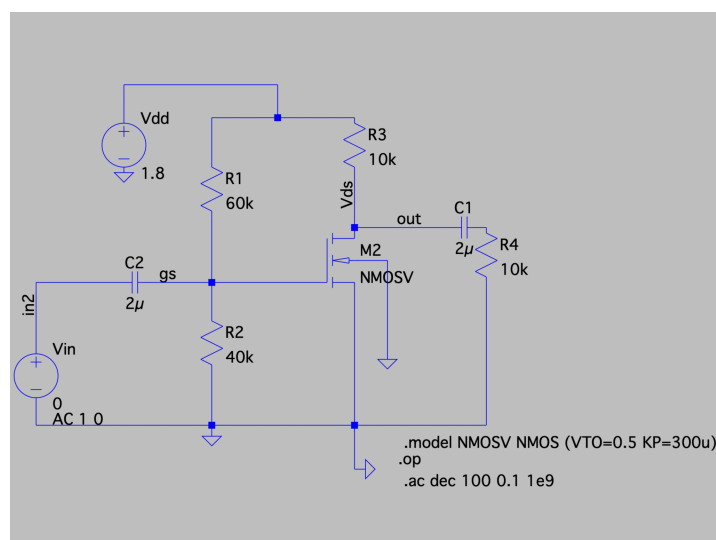
The small-signal transconductance g_m was approximately 0.6–0.7 mS.

The mid-band voltage gain was estimated using:

$$A_v \approx -g_m \times R_D$$

which resulted in a gain close to 10 dB.

- Circuit diagram:-



1.DC Biasing and Operating Point Analysis:-

DC analysis was performed to determine a suitable bias point for the NMOS transistor and to ensure operation in the saturation region.

A DC sweep of the gate bias voltage was first carried out to observe the variation of drain current with input bias.

This sweep helped identify the voltage range that produced the desired operating current. Following this, an operating point (.op) analysis was executed to extract node voltages and device parameters.

The final bias condition resulted in a gate-to-source voltage of approximately 0.72 V and a drain-to-source voltage of approximately 1.07 V.

The corresponding drain current stabilized in the range of 70–75 μA .

These values confirmed that the MOSFET was biased in the saturation region, satisfying the condition:

$$V_{DS} > V_{GS} - V_{TH}$$

Bias resistor values and device dimensions were iteratively adjusted until the required operating point was achieved with sufficient margin from the triode region.

This DC operating point formed the foundation for subsequent small-signal, AC, and noise analyses

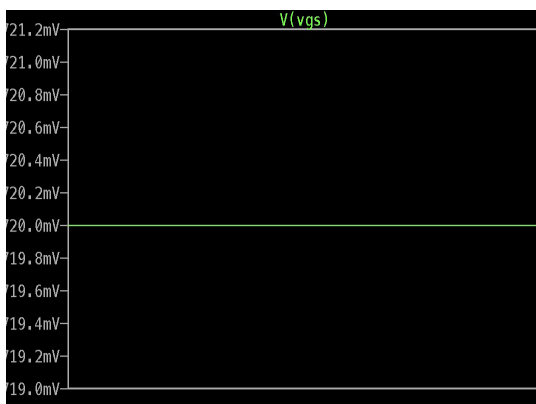
Specifications:-

```
vin: Missing value, assumed 0V @ DC
Direct Newton iteration for .op point succeeded.
Semiconductor Device Operating Points:
----- MOSFET Transistors -----
Name:      m2
Model:     nmosv
Id:        7.26e-05
Vgs:       7.20e-01
Vds:       1.07e+00
Vbs:       0.00e+00
Vth:       5.00e-01
Vdsat:     2.20e-01
Gm:        6.60e-04
Gds:       0.00e+00
Gmb:       0.00e+00
Cbd:       0.00e+00
Cbs:       0.00e+00
Cgssov:    0.00e+00
Cgdov:     0.00e+00
Cgbov:     0.00e+00
Cgs:       0.00e+00
Cgd:       0.00e+00
Cgb:       0.00e+00
```

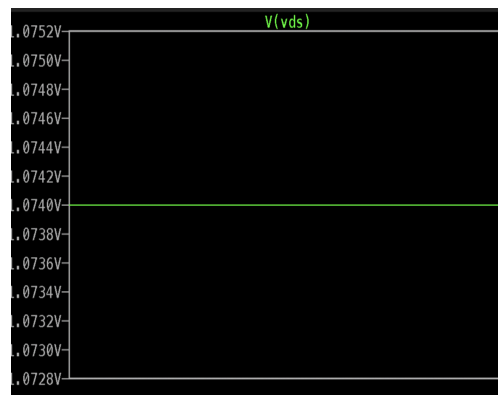
```
Operating Bias Point Solution:
V(n001)      1.8      voltage
V(vgs)       0.72     voltage
V(p001)      2.148e-14 voltage
V(vds)       1.074     voltage
V(n002)       0       voltage
Id(M2)       7.26e-05  device_current
Ig(M2)       0         device_current
Ib(M2)      -1.084e-12 device_current
Is(M2)      -7.26e-05  device_current
I(C1)       -2.148e-18 device_current
I(C2)       1.44e-18   device_current
I(R1)       1.8e-05     device_current
I(R2)       1.8e-05     device_current
I(R4)       2.148e-18   device_current
I(R3)       7.26e-05   device_current
I(Vdd)      -9.06e-05   device_current
I(Vin)      1.44e-18   device_current

Total elapsed time: 0.029 seconds.
```

Vgs:-



Vds:-



2.AC Frequency Response Analysis:-

After finalizing the DC biasing and device sizing, AC small-signal analysis was performed to evaluate the frequency-domain behavior of the amplifier.

An AC simulation was run by applying a small-signal sinusoidal input source while sweeping frequency over a wide range.

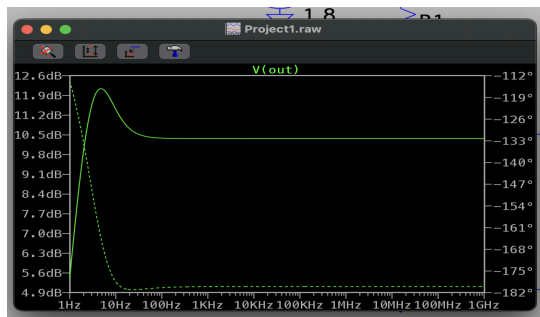
The resulting Bode magnitude and phase plots were used to study the gain profile across frequency.

The mid-band voltage gain was verified to be approximately 10 dB, confirming agreement with theoretical predictions from the small-signal model.

The lower -3 dB cutoff frequency was extracted from the magnitude response and found to be approximately 1.2 Hz, primarily determined by the input and output coupling capacitors.

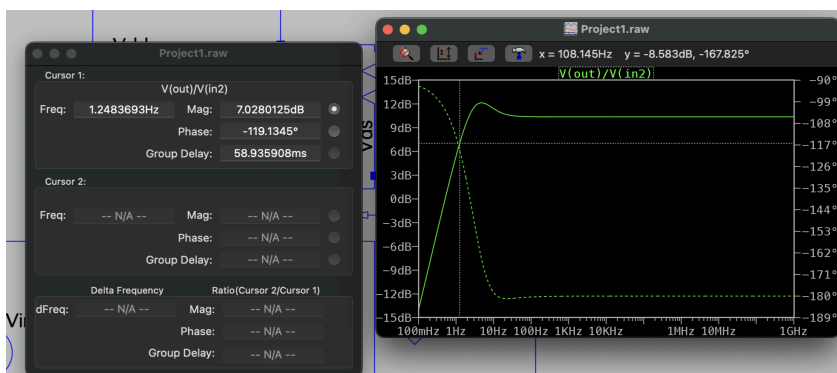
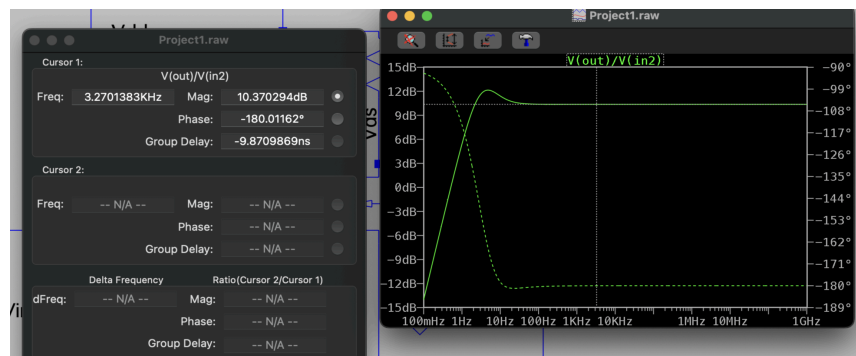
At high frequencies, the gain remained flat well into the GHz range, with the upper -3 dB cutoff frequency exceeding 1 GHz, indicating wideband operation limited mainly by device parasitics.

This analysis validated that the amplifier met the targeted bandwidth specifications and was suitable for small-signal amplification over a broad frequency range.



- Vout

- Mid band gain



- -3db Gain

3. Transient Response Phase :-

Transient analysis was performed to study the time-domain behavior of the CMOS common-source amplifier and to verify proper amplification of dynamic input signals.

A small-amplitude sinusoidal voltage source was applied at the input through the coupling capacitor while observing the output waveform at the drain node.

The simulation was run for sufficient time duration to allow initial transients to settle and for steady-state operation to be achieved.

The output waveform exhibited clear amplification relative to the input signal along with a 180° phase inversion, which is characteristic of a common-source topology.

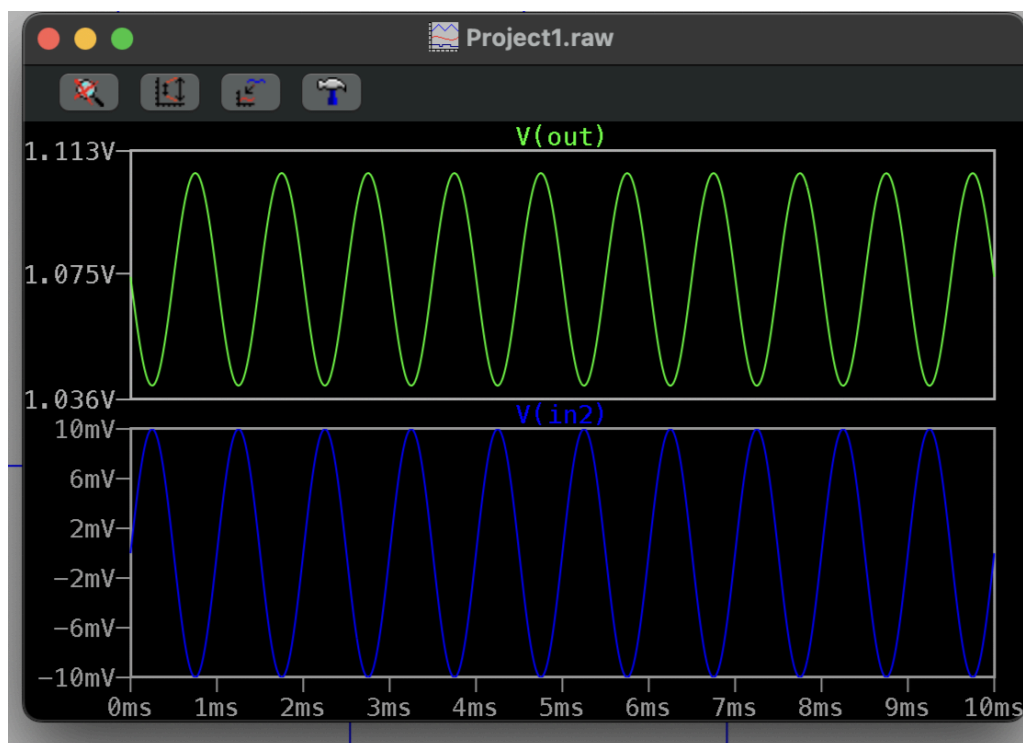
No clipping or waveform distortion was observed at the selected bias point, confirming that the transistor remained in the saturation region during signal swings.

Rise time, fall time, and steady-state amplitude were inspected to ensure that the circuit responded quickly and linearly to changes in the input signal.

The results verified that the amplifier could faithfully reproduce time-varying inputs within the designed small-signal range.

This transient-domain verification complemented the DC and AC analyses and confirmed that the circuit operates correctly under real-signal excitation conditions.

- Transient inversion:-



4.Parameter Sweep and Design Optimization:-

Parametric sweep analysis was carried out to study the sensitivity of the CMOS common-source amplifier to key design variables and to optimize performance.

The NMOS transistor width W was swept while keeping the channel length constant in order to observe its effect on drain current, transconductance, and mid-band gain.

Increasing W resulted in higher drain current and transconductance, which in turn increased the voltage gain, while excessive widening pushed the operating point closer to the edge of saturation.

An optimum value of W was therefore selected to balance gain, power consumption, and bias stability.

The gate-bias resistor $R3$ was swept to examine its influence on the gate voltage and DC operating point.

Variation of $R3$ shifted the gate bias level, directly affecting V_{GS} and drain current.

This sweep was used to fine-tune the bias point so that the transistor consistently operated in the saturation region.

The drain resistor $R4$ was swept to evaluate its effect on output voltage swing and voltage gain.

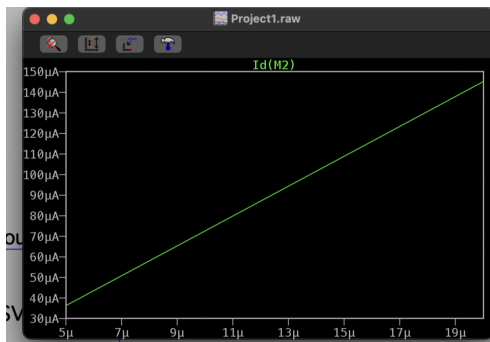
Increasing $R4$ raised the small-signal gain but reduced the available output headroom, while smaller values lowered gain but improved linear swing.

An optimized $R4$ value was selected to achieve the targeted mid-band gain of approximately 10 dB while maintaining sufficient V_{DS} .

The combined sweep results were used to choose final component values that satisfied DC biasing requirements, small-signal gain targets, and bandwidth constraints simultaneously.

This phase demonstrated a systematic optimization process using SPICE-based sensitivity analysis rather than trial-and-error tuning.

W sweep :-



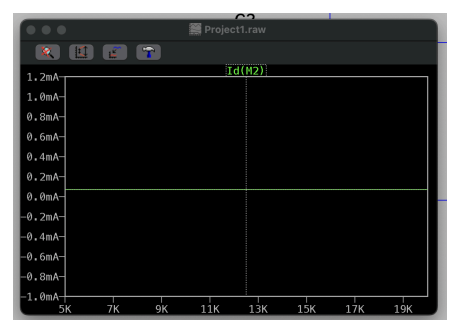
```
.model NMOSV NMOS (VTO=0.5 KP=300u)
.param W=10u
.op
.step param W list 5u 10u 20u
```

R3 sweep :-



```
.model NMOSV NMOS (VTO=0.5 KP=300u)
.param R3=10k
.op
.step param R3 list 5k 10k 20k 40k
```

R4 sweep:-



```
.model NMOSV NMOS (VTO=0.5 KP=300u)
.param R3=10k
.op
.step param R3 list 5k 10k 20k 40k
```

5.Process Corners and Temperature Analysis:-

Process corner and temperature simulations were performed to evaluate the robustness of the CMOS common-source amplifier under manufacturing and environmental variations. Typical, fast, and slow device conditions were considered to represent variations in transistor threshold voltage, mobility, and parasitic capacitances.

These corners were used to observe the resulting shifts in DC operating point, gain, and bandwidth.

Across the simulated corners, the amplifier continued to operate in the saturation region, although moderate variations in drain current and transconductance were observed.

The mid-band gain remained close to the nominal 10 dB target, confirming acceptable tolerance to process spread.

Temperature sweeps were carried out over a wide range to study thermal effects on circuit behavior.

As temperature increased, a reduction in carrier mobility caused a slight decrease in transconductance and gain, while threshold voltage variations produced corresponding changes in bias current.

At lower temperatures, the opposite trend was observed.

Despite these changes, the circuit maintained correct functionality and remained biased in saturation over the full temperature range examined.

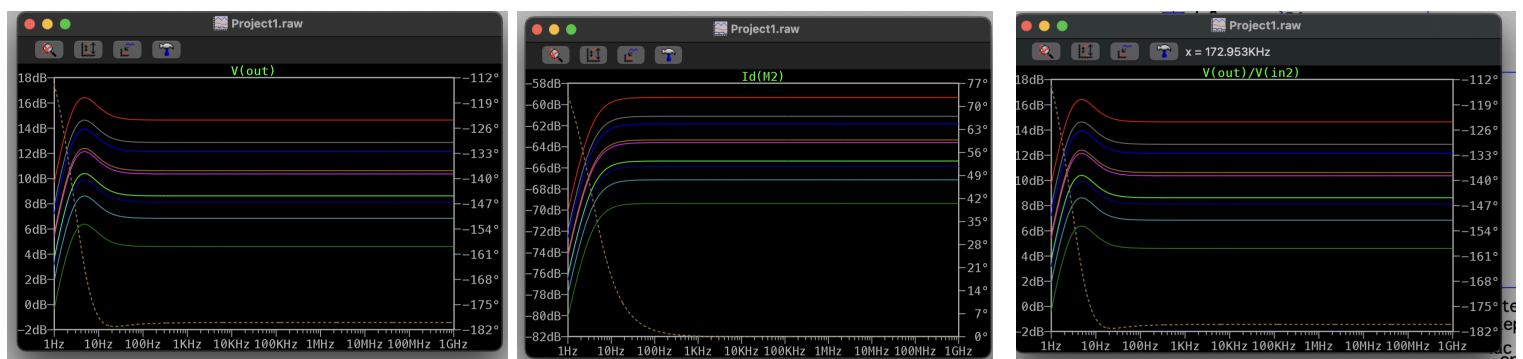
This analysis demonstrated that the design possesses adequate robustness for practical integrated-circuit environments.

- Temperature:-



```
.model NMOSV NMOS (VTO=0.5 KP=300u)
.temp -40 27 125
.ac dec 100 1 1e9
.op
```

- Corners:-



```
.param KP=300u
.param VTO=0.5
.model NMOSV NMOS (VTO={VTO} KP={KP})
```

```
.step param VTO list 0.45 0.5 0.55
.step param KP list 200u 300u 400u
.ac dec 100 1 1e9
.op
```

6. Noise Performance Analysis:-

Noise analysis was performed to evaluate the small-signal noise behavior of the CMOS common-source amplifier across frequency.

A SPICE noise simulation was executed to compute the output-referred noise spectral density while referencing the input source.

Both thermal noise from resistive elements and channel noise from the MOSFET were captured in the analysis.

At low frequencies, the noise spectrum was dominated by flicker ($1/f$) noise contributions from the MOS device, while at mid and high frequencies thermal noise components became more significant.

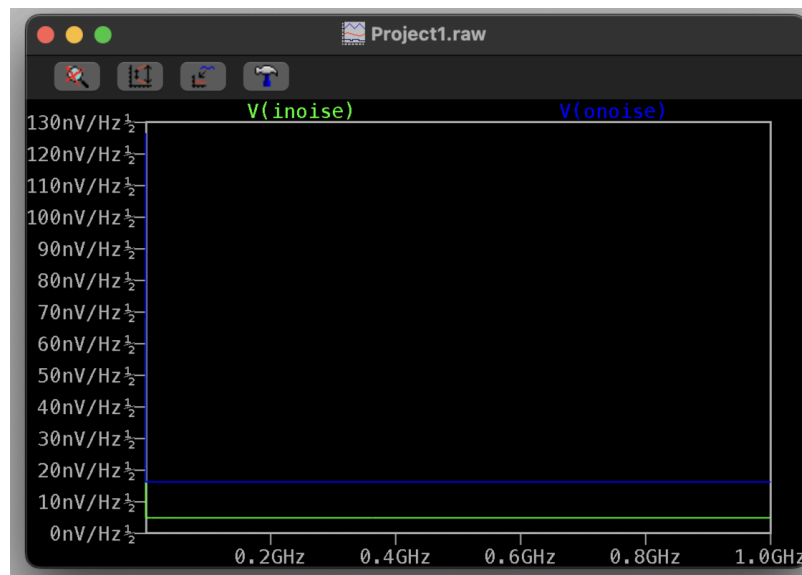
The overall noise profile remained smooth and well-behaved over the operating bandwidth of the amplifier.

The effect of device sizing and biasing on noise was also observed during earlier parametric sweeps.

Increasing transistor width reduced flicker-noise density but increased parasitic capacitances, leading to trade-offs between noise performance and bandwidth.

The noise results confirmed that the selected bias point and device dimensions provided acceptable noise characteristics for small-signal analog amplification.

This phase completed the frequency-domain verification of the amplifier and strengthened confidence in the circuit's suitability for integrated-circuit applications.



RESULT :-

This project presents the phase-wise design and simulation of a CMOS common-source amplifier using LTspice.

Starting from specification definition and topology selection, the circuit was progressively refined through DC biasing, device sizing, transient response analysis, AC frequency response, noise evaluation, parameter sweeps, and corner and temperature verification.

The final design achieved stable saturation-region operation with a mid-band gain of approximately 10 dB and wideband frequency response extending beyond 1 GHz.

Robustness against process and temperature variations was also demonstrated, highlighting the applicability of the design for analog and mixed-signal integrated circuits.

<u>Parameter.</u>	<u>Final Obtained Value</u>
• Gate-to-Source Voltage (VGS)	$\approx 0.72\text{ V}$
• Drain-to-Source Voltage (VDS)	$\approx 1.07\text{ V}$
• Drain Current (ID)	70–75 μA
• Transconductance (gm)	0.6–0.7 mS
• Mid-Band Gain	$\approx 10\text{ dB}$
• Lower -3 dB Frequency (fL)	$\approx 1.2\text{ Hz}$
• Upper -3 dB Frequency (fH)	$> 1\text{ GHz}$
• Operating Region	Saturation
• Corner & Temp Stability.	Maintained saturation

Conclusion:-

A CMOS common-source amplifier was successfully designed and verified using a structured phase-wise engineering approach in LTspice.

Through iterative optimization of biasing networks, transistor sizing, and load resistance, stable saturation-region operation was achieved while meeting the targeted gain and bandwidth requirements.

Transient, AC, and noise simulations confirmed correct time- and frequency-domain behavior, while parameter sweeps and corner-temperature analyses demonstrated design robustness against realistic process and environmental variations.

This project strengthened fundamental analog CMOS design skills and reflects a professional workflow relevant to analog and mixed-signal VLSI roles.